

Research plan: estimating the disease burden of food contamination in India from open data

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Live platform: <https://foodsafev2.vercel.app> · Code and data pipeline:
<https://github.com/ultramagnus23/FoodSafe>

This document is the scope of the paper: what it asks, what it will and will not include, which data it uses, how each part is done and checked, and the timeline. It supersedes the paper scope in `PAPER_SCOPING.md` (the data-access audit there becomes one part of this paper; see section 14).

1. Summary

Food contamination causes disease, but in India nobody can say how much, where, or from what. The food regulator publishes no machine-readable test results, and no public dataset records where contaminated food makes people ill. A disease "predictor" trained on contamination and illness data is therefore not possible: there is nothing to train it on, and building one would mean inventing ground truth.

There is a defensible alternative, and it is the one WHO and JECFA use: quantitative risk assessment. The dose-response relationship comes from toxicology and epidemiology that already exist; exposure is estimated from how contaminated foods are and how much of them people eat; expected cases follow. FoodSafe already holds most of the pieces as an open, cited, daily-updated platform: India's legal limits beside the EU, Codex and the US (56,619 limits, 1,711 compared pairs), 32,805 EU contamination findings across 164 origin countries, India's state testing outcomes from Parliament, a hazard-to-disease knowledge base with a source for every link, and a validated hazard classifier.

The paper builds the missing step for one hazard where the science is strongest, aflatoxin and liver cancer, and asks four things: how India's limits compare and where the gaps matter for health; what annual burden of liver cancer dietary aflatoxin implies for India, by state, using only public data; whether that estimate agrees with independent estimates; and which missing data limit it most. The engine code for the burden step is written and already reproduces a published risk assessment for India exactly (section 8.3).

2. Why this matters

- WHO estimates 1.52 million deaths a year worldwide from foodborne disease (2021 estimates, loaded in the platform). Among chemical hazards, inorganic arsenic (641,000 deaths) and lead (466,000) lead; aflatoxin B1 causes an estimated 13,996 liver cancers and 13,175 deaths a year (412,396 DALYs). WHO publishes these at global level only.
- India had about 30,000 new liver cancers in 2016, and the male rate varied 7.9-fold across states (India State-Level Disease Burden Initiative, Lancet Oncology 2018). How much of that is dietary aflatoxin is unknown. Liu & Wu (2010) put India's aflatoxin exposure somewhere between 4 and 100 ng/kg body weight per day, a 25-fold range, because the occurrence data were thin.
- India's limits are often looser than the EU's. Aflatoxin M1 in milk: 0.5 µg/kg in India, 0.05 in the EU. Total aflatoxins in spices: 30 µg/kg in India, 10 in the EU. Whether that matters for health is a question nobody has put a number on.

- India's regulator blocks anonymous access to its testing and recall portal. The only real pass/fail data over 13 years is state-level yearly totals disclosed to Parliament. A method that works on public data, and says exactly what data would improve it, is useful to researchers, journalists and the regulator itself.

3. Research questions and hypotheses

	Question	Pre-specified hypothesis or expectation
RQ1 Rules	How do India's legal limits for contaminants and pesticide residues compare with the EU, Codex and the US, and which gaps involve hazards with serious health outcomes?	H1: India's limit is above the EU's in most compared pesticide pairs, mostly because the EU permits no use of that pesticide on that food (as measured on 2026-10-03: 759 of 1,297 pesticide pairs higher; 683 of those are EU detection-floor limits). Re-measured on the frozen snapshot.
RQ2 Hazard identification	Can public contamination records be classified by hazard reliably enough to feed the engine?	Already measured: accuracy 0.918 and macro-F1 0.777 on 6,934 later notifications, against 0.796 / 0.586 for a keyword baseline; 0.947 on India-origin notifications. The paper adds calibration and an error analysis.
RQ3 Burden (primary)	Using only public data, what annual burden of liver cancer does dietary aflatoxin imply for India, nationally and by state, and does it agree with independent estimates?	H2: the national estimate falls inside the range implied by Liu & Wu (2010) for India, is a small fraction of WHO's 2021 global aflatoxin total, and in every state stays below total liver-cancer incidence (a hard plausibility bound; see 8.3).
RQ4 Scenario	How would the estimate change if India's aflatoxin limits were the EU's or Codex's?	Expectation (not a hypothesis test): the change is concentrated in the foods where India's limit is furthest above the EU's (milk, spices); its size depends on how much of the occurrence distribution lies between the two limits.
RQ5 Data gaps	Which missing data limit the estimate most, and what would it take to go from state to district resolution?	H3: uncertainty in contamination levels contributes more to output variance than consumption, hepatitis B prevalence or potency.

4. Contributions

- An open disease-risk engine for food contamination** built only from public records, following the Codex risk-assessment steps, with every value traceable to its source, a document hash and a confidence level. Software, not a one-off analysis: it updates daily.
- The first machine-readable, health-linked comparison of India's food limits** with the EU, Codex and the US, with the reason each gap exists.
- An open, reproducible state-level estimate of aflatoxin-attributable liver cancer in India**, using current consumption data (HCES 2022-23), with stated uncertainty and checked against independent estimates.
- A quantified account of the data gap**: which inputs drive the uncertainty, and dated evidence of what India's regulator does and does not make available.

5. Scope

In scope

Area	Included
Geography	India: national and state/UT level. Other countries only as benchmarks and context.

Area	Included
Burden estimate	Aflatoxins (B1 and its total; M1 in milk) and hepatocellular carcinoma (ICD-10 C22.o). Foods: groundnut, maize, rice, chillies and other spices, milk; others if the review finds Indian occurrence data.
Screening	Hazard quotient or margin of exposure for every hazard with a measured concentration on Indian food (EU findings and the reviewed literature), as a risk flag, not a case count.
Standards	All 1,711 India-vs-world pairs from the frozen snapshot (pesticide MRLs and contaminant maximum levels).
Hazard identification	The existing classifier: evaluation, calibration, errors.
Data gaps	Variance decomposition of the burden estimate, plus the dated access evidence (FoSCoS probe log, RTI reply, extraction cost of the parliamentary data).

Out of scope, and why

Excluded	Why
A machine-learning model trained to predict disease cases from contamination	No dataset pairs contamination and illness in the same place. Training one would require inventing outcomes. The engine predicts expected burden through published dose-response instead.
District, city or market estimates	No public contamination data exists at that grain (section 7.3). The paper says what would be needed.
Microbial hazards (Salmonella, E. coli, etc.)	Their burden is estimated from incidence surveillance, not dose-response; India's outbreak surveillance (IDSP) reports were not reachable. Kept as descriptive context.
Lead, inorganic arsenic, other chemicals	Their dose-response runs through blood lead or water exposure and needs different inputs. Named as the next extension; screened only.
Prevalence of contamination	Every source is risk-targeted testing (border checks, inspector sampling). No prevalence claims.
Ratings of brands or producers; health advice	Not what the data support, and legally and ethically out of bounds.
Causal or ecological correlation claims ("state X has more cancer because of food")	The estimate is a model output, checked for plausibility, not an observed association.
Nutrition of packaged food; the global susceptibility index	On the platform, but separate papers.
Any data obtained by getting around access controls	The audit is read-only by design; no captcha or login was bypassed.
Synthetic or "demo" data	Never used. The platform's production database is real-only (decided 2026-09-11).

6. Framework: the engine, step by step

The engine follows the four steps of food-safety risk assessment in the Codex Alimentarius procedural manual, plus a validation step.

Step	What it answers	Engine component	Status
Context: rules	What each country's law allows	<code>standards_*.py</code> , <code>models/standards_compare.py</code>	Built

Step	What it answers	Engine component	Status
Hazard identification	Which hazard a finding or report is about	models/health_classifier.py (knowledge-base rules), models/hazard_text_classifier.py (ML)	Built and validated
Hazard characterisation	What the hazard does, to whom, at what dose	pipeline/sources/hazard_kb.py (52 curated hazards, 5 pesticide classes, IARC groups for 1,128 agents, cited outcomes); hazard_reference_values (ADI, ARfD, TDI); cancer potency in models/burden_engine.py	Built
Exposure assessment	How much people take in	models/safe_intake.py (single findings); models/burden_engine.py (population exposure from occurrence x consumption / body weight)	Code built; India inputs to assemble (section 7.2)
Risk characterisation	Expected cases, hazard quotient, margin of exposure	models/burden_engine.py (cases with Monte Carlo uncertainty, MOE, HQ)	Code built and tested
Validation	Does the estimate agree with independent evidence?	Section 8.3	Planned

7. Data

7.1 In hand (production database, refreshed daily or weekly)

Dataset	Size	Period	Grain	Confidence	Role in the paper
Legal limits: FSSAI, EU, Codex, US	56,619 limits; 1,711 India-vs-world pairs	current rule-books	food x hazard	high (EU, Codex, US), medium (FSSAI, PDF with integrity checks)	RQ1; limits for RQ4
EU RASFF notifications, all origins	32,805 notifications, 164 origins; India 2,228 (3rd)	2019 to now	consignment x hazard, with measured value and limit	high	RQ2 training/testing; measured values for screening; upper-bound occurrence for RQ3 sensitivity
Lok Sabha state sampling outcomes	539 state-year rows from 16 validated tables	2013-14 to 2025-26	state x year	medium	Context; RQ5 evidence (extraction cost)
Pesticide-residue monitoring (MPRNL)	73 rows	2012-13 to 2018-19	national x commodity	medium	Screening context
Hazard-to-health knowledge base	52 curated hazards with 70 cited outcome links (789 hazard-outcome rows in all), 5 pesticide classes, IARC groups for 1,128 agents	current	hazard x outcome	medium (curated, cited)	Hazard characterisation
Reference values (ADI, ARfD, TDI)	EFSA, JMPR, JECFA	current	substance	high	Screening

Dataset	Size	Period	Grain	Confidence	Role in the paper
WHO foodborne burden (FERG 2021)	288 rows	2021	global x hazard x age	high	Benchmark for RQ3
Country indicators (WHO, World Bank)	217 countries	latest years	country	high	Context
Peer-reviewed literature (OpenAlex, Europe PMC)	2,569 papers on 9 contaminants	to now	paper	medium (retrieved, not vetted)	Starting corpus for the occurrence review (RQ3)
FoSCoS access probe log	15 dated readings	2026-07-04 to now	reading	n/a	RQ5 evidence

7.2 To acquire (all public; access terms checked in week 1)

Input	Source	Access	Effort	Risk
Aflatoxin occurrence in Indian foods	Systematic review of published surveys (PRISMA 2020): the 2,569-paper corpus plus PubMed, Scopus and Google Scholar searches	open literature	3 weeks, the largest task	Uneven regional coverage; heterogeneous methods; publication bias
Aflatoxin M1 in milk	FSSAI National Milk Safety and Quality Survey 2018 report; a Lok Sabha answer gives 368 of 6,432 samples above the limit	public PDF	2 days	Only the share above the limit may be published, not the distribution
Food consumption by state	MoSPI Household Consumption Expenditure Survey 2022-23 (and 2023-24): item-level quantities, rural and urban	public reports; unit-level data via MoSPI's microdata portal (registration)	1 week	Items may be grouped (e.g. groundnut vs groundnut oil); food eaten outside the home under-recorded
Body weight	ICMR-NIN reference body weights (2020); NFHS-5 state anthropometry for sensitivity	public; NFHS microdata via DHS Program registration	2 days	
Hepatitis B prevalence	Published meta-analyses (general population roughly 1.5-3.7%) and the 2017-18 national serosurvey (1.1% in children 5-17)	open literature	2 days	Few adult state-level estimates; handled as a range
Population by state	MoHFW Technical Group population projections	public	1 day	
Liver-cancer incidence by state (validation)	GBD 2021 India subnational results (IHME Results Tool); NCRP registry reports (ICMR-NCDIR)	free registration; non-commercial licence	2 days	Modelled estimates themselves; used as a bound, not ground truth
Potency and benchmarks	JECFA potencies via EFSA (2020), verified; Liu & Wu (2010), verified	open	done	

7.3 Not available, with the evidence

- **District or market contamination testing.** FSSAI's FoSCoS portal refuses anonymous access (401 from a local network; no connection at all from a GitHub runner); 15 dated readings in

docs/foscoc_access_log.jsonl. Enforcement-report URLs redirect to the homepage (docs/FSSAI_INGESTION.md).

- **Outbreak and illness data by place.** IDSP outbreak reports were unreachable; NCDIR cancer data is registry-level with no export.
- **Through RTI.** Filed 2026-07-11 (FSSAI/R/E/26/00836); answered as unable to provide the data.
- **Through Parliament.** Recoverable, but at state-year grain only and at high cost: of 70 candidate tables in 124 answers, 54 failed extraction checks.

8. Methods

8.0 Common rules

- **Freeze.** The analysis runs on a dated database snapshot (dump plus sha256), so every number can be regenerated. The live platform keeps updating; the paper does not.
- **Pre-registration.** This plan's hypotheses, decision rules and analyses are time-stamped on OSF before the burden model is run on India data.
- **No tuning to targets.** Inputs are fixed from the data before comparison with benchmarks. If a validation check fails, the paper reports the failure and its cause; it does not adjust inputs until the check passes.

8.1 Study 1: rules (RQ1)

Descriptive analysis of the 1,711 pairs: counts and ratios by jurisdiction, by type (pesticide MRL or contaminant ML), and by the reason for each gap (EU not approved, no use on that food, never assessed, EU level above detection). Health-linking: each pair is joined to the knowledge base, and pairs where India's limit is higher and the hazard is IARC Group 1 or 2A, or has a cited serious outcome, are reported separately. Sensitivity: exclude pairs where India's value comes from a group or catch-all limit. Framing rule: a limit is a regulatory line, not a safety verdict.

8.2 Study 2: hazard identification (RQ2)

The classifier is already trained on 15,210 EU notification subjects (2020-2024) and tested on the 6,934 that came later (2025-2026). Added for the paper: per-class precision and recall, a reliability diagram and expected calibration error, a McNemar test against the keyword baseline, and a hand-labelled error analysis of 100 India-origin misclassifications. It classifies text; it detects nothing.

8.3 Study 3: aflatoxin and liver cancer (RQ3), the core

Exposure. For state s :

$$E_s = \text{sum over foods } f \text{ of } C_f \times I(f,s) / BW_s \quad (\text{ng per kg body weight per day})$$

where C_f is the aflatoxin concentration distribution in food f ($\mu\text{g/kg}$), $I(f,s)$ the mean daily intake of f in state s (g/day, from HCES), and BW_s body weight. Aflatoxin M1 is weighted at 0.1 of B1's potency (EFSA 2020).

- *Occurrence (C_f)* from the systematic review: per food, pool reported means, SDs and percentiles into a lognormal distribution. Non-detects handled by EFSA practice (lower bound = 0, upper bound = LOD), reported both ways. RASFF values on Indian exports are an export-biased upper bound, used only as a sensitivity case.

- *Consumption (I)* from HCES 2022-23 state means, rural and urban, converted from monthly per-capita quantities to g/day.
- *Susceptibility*: hepatitis B carrier share as a range (section 7.2), higher in the north-eastern states where data show it.

Risk. Expected incidence per 100,000 per year:

$$\text{incidence_s} = E_s \times (\text{potency_HBsAg+} \times p_s + \text{potency_HBsAg-} \times (1 - p_s))$$

using JECFA 2016 potencies (0.269 and 0.017 per 100,000 per year per ng/kg bw/day; upper bounds 0.562 and 0.049), and cases = incidence x population / 100,000. Margin of exposure against EFSA's BMDL10 of 0.4 µg/kg bw/day is reported alongside (below 10,000 = concern). DALYs follow FERG: cases converted with GBD 2021 disability weights and life expectancy at onset.

Uncertainty. Monte Carlo, 10,000 draws over concentration, intake, body weight, carrier share and potency (uniform between central and upper bound); median and 95% interval.

Validation, in four checks, decided before running:

1. *Replication (done).* From Liu & Wu's inputs for India (exposure 4 to 100 ng/kg bw/day, JECFA 1998 potencies), the engine returns their published incidences exactly: 0.04 to 1.00 per 100,000 for non-carriers and 1.20 to 30.0 for carriers (tests/test_burden_engine.py).
2. *External consistency.* The national estimate is compared with the range implied by Liu & Wu for India and with WHO's 2021 global aflatoxin total (13,996 cases). India's estimate must be a plausible fraction of the global figure.
3. *Plausibility bound.* In every state, attributable cases must stay below GBD's total liver-cancer incidence; the implied attributable fraction is compared with published ranges (Liu & Wu: 4.6% to 28.2% of liver cancers worldwide). A violation means the inputs or model are wrong, and is reported as such.
4. *Sensitivity.* One-at-a-time and variance-based (Sobol) sensitivity identify which inputs drive the result (feeds RQ5).

8.4 Study 4: limit scenarios (RQ4)

Following EFSA's assessment of raising the EU peanut limit from 4 to 10 µg/kg, concentrations above a limit are set to that limit (full compliance), first at India's limit, then at the EU's and Codex's, and the difference in expected cases is reported. This shows how much of the estimated burden sits in the band between the limits. It is a scenario, not a forecast: compliance and enforcement are not modelled.

8.5 Study 5: data gaps (RQ5)

From 8.3's variance decomposition: how much narrower the interval would be if each input's uncertainty were halved (a simple value-of-information measure). Combined with the dated access evidence (section 7.3) into concrete recommendations: what the regulator would need to publish (machine-readable results with food, hazard, value, limit, district, date) and what a minimal sampling study would need to measure.

8.6 Exploratory, only if time allows (not in the core paper)

Country-level check: does open contamination evidence (RASFF hazard profiles) add information about national foodborne burden beyond development indicators? Nested cross-validated models; a null result is expected and would be reported as one.

9. Validation at a glance

Component	Check	Pass criterion	Status
Standards extraction	Integrity checks per rule-book; spot checks against printed documents	Documented in docs/STANDARDS.md	Done
Hazard classifier	Temporal hold-out, baseline, calibration	Beats baseline (McNemar $p < 0.05$)	Accuracy and baseline done
Burden arithmetic	Reproduce Liu & Wu (2010) India incidences	Exact	Done (19 tests)
Burden estimate	Consistency with Liu & Wu and FERG 2021	Inside the implied range	Planned
Burden estimate	Below GBD total liver-cancer incidence in every state	No violation	Planned
Inputs	Double extraction on 20% of reviewed studies	Agreement reported (kappa / ICC)	Planned

10. What the paper will not claim

- That anyone was harmed by a specific food, brand or place.
- That contamination is common or rare in India (all sources are targeted testing).
- That a higher Indian limit makes Indian food unsafe; only what the difference implies under stated assumptions.
- That the burden estimate is an observed count. It is a model output with an interval.
- That the classifier tests food.

11. Ethics and data governance

- Public, aggregate data only; no personal data. The one named person RASFF can carry (a notifier's contact) is never stored.
- Read-only access throughout. The FoSCoS audit records what a good-faith visitor can reach and never attempts to get around the captcha or login.
- Licences respected: GBD and DHS data under their user agreements (not redistributed); MoSPI unit-level data under its terms; derived results released with provenance.
- Harm: nothing names producers; results are framed so as not to alarm, with limits stated next to each figure.

12. Limitations and threats to validity

- Literature occurrence data are convenience samples with uneven regional coverage and mixed methods; publication may favour high findings. Mitigation: quality scoring, sensitivity excluding low-quality studies, bounds for non-detects.
- HCES records household purchases, grouped items and little food eaten outside the home.
- Potency comes from cohorts in China and Africa; transfer to India is an assumption, stated.
- Liver cancer has other causes (hepatitis C, alcohol, metabolic disease); the engine estimates the aflatoxin share only, checked against the total.
- Cancer appears years after exposure; the model assumes steady state.
- Single country, single hazard for the burden estimate.

13. Timeline (12 weeks from supervisor agreement)

Week	Work	Milestone
1	Freeze snapshot; confirm data access and licences; OSF pre-registration; supervisor review of this plan	Protocol registered
2-4	Systematic review of aflatoxin occurrence: search, screening, extraction, double extraction on 20%	Occurrence dataset
3-4	In parallel: consumption, body weight, carrier share, population, GBD validation data	Input tables
5	Burden runs, validation checks 2-4	Results, pass or fail
6	Scenarios, variance decomposition; final standards and classifier analyses	All figures
7-10	Writing; internal review	Full draft
11	Supervisor review; revisions	Submission draft
12	Preprint and journal submission; code and data release with DOI	Submitted

14. Relation to the earlier scope

PAPER_SCOPING.md committed to an audit of India's food-safety data access. That evidence is not dropped: it is RQ5 and the paper's explanation of why the burden estimate stops at state level. If a supervisor prefers, it can also stand alone as a short companion paper for a civic-tech venue (COMPASS / AI for Social Good).

15. Deliverables and venues

- Paper. Preprint on medRxiv (public health) with the full method; journal candidates: *Food and Chemical Toxicology*, *Food Control*, *Risk Analysis*, or *Toxins* for the aflatoxin focus. The supervisor's view decides.
- Code (open source) and the frozen dataset with a DOI (Zenodo), including the occurrence dataset from the review.
- The live platform, with the engine page showing the burden step once validated.

16. What I am asking a supervisor for

1. Is the risk-assessment framing (expected burden from published dose-response) the right way to build a disease engine without paired outcome data? What would make it convincing?
2. Review of the occurrence-review protocol and the validation checks before the model runs.
3. Advice on which journal, and whether the data-access audit belongs inside or beside it.
4. Introductions: epidemiologists or toxicologists working on aflatoxin, liver cancer or dietary exposure in India; anyone with access to state-level testing data.
5. If appropriate: supervision of the paper and co-authorship as the contribution warrants.

17. Status on 2026-10-06

Piece	State
Platform (standards, findings, hazards, places, nutrition, sources, engine page)	Live at https://foodsafev2.vercel.app ; data refreshed daily by CI

Piece	State
Standards comparison, RASFF all origins, knowledge base, classifier	Built, validated, in production
Burden engine (models/burden_engine.py)	Written; reproduces Liu & Wu (2010) for India; 19 tests
India inputs for the burden estimate	Not yet assembled (section 7.2)
Pre-registration, review, writing	Not started
Known issue	The July 2026 district engine (models/disease_burden.py, dose_response_params seeds in schema_migration_003.sql) cites "IARC Monograph" for aflatoxin slope factors, but IARC publishes no slope factors, and the seeded values differ from JECFA's by roughly 10-20 times. It produces no output on real data (0 estimates) and is not used by this plan; the seeds should be corrected or retired.

References (all checked)

- EFSA CONTAM Panel (2020). Risk assessment of aflatoxins in food. *EFSA Journal* 18(3):6040. <https://doi.org/10.2903/j.efsa.2020.6040> (JECFA 2016 potencies; AFM1 potency factor 0.1; BMDL10 0.4 µg/kg bw/day; MOE below 10,000 a concern.)
- EFSA CONTAM Panel (2018). Effect on public health of a possible increase of the maximum level for "aflatoxin total" from 4 to 10 µg/kg in peanuts. *EFSA Journal* 16(2):e05175. <https://doi.org/10.2903/j.efsa.2018.5175> (limit-scenario method.)
- Liu Y, Wu F (2010). Global burden of aflatoxin-induced hepatocellular carcinoma: a risk assessment. *Environmental Health Perspectives* 118(6):818-824. <https://doi.org/10.1289/ehp.0901388> (25,200-155,000 cases a year worldwide; 4.6-28.2% of liver cancers; India exposure 4-100 ng/kg bw/day.)
- India State-Level Disease Burden Initiative Cancer Collaborators (2018). The burden of cancers and their variations across the states of India: the Global Burden of Disease Study 1990-2016. *Lancet Oncology* 19(10):1289-1306. [https://doi.org/10.1016/S1470-2045\(18\)30447-9](https://doi.org/10.1016/S1470-2045(18)30447-9) (30,000 liver cancers in 2016, 95% UI 29,000-32,000; crude male incidence varied 7.9-fold across states.)
- Hepatitis-B virus infection in India: findings from a nationally representative serosurvey, 2017-18. *International Journal of Infectious Diseases* (2020). <https://www.sciencedirect.com/science/article/pii/S1201971220307128> (HBsAg 1.1% in children 5-17.)
- WHO Foodborne Disease Burden Epidemiology Reference Group, 2021 estimates, as loaded in the platform (foodborne_burden_global).
- Platform method documents: docs/STANDARDS.md, docs/HAZARD_KB.md, docs/MODEL_HAZARD_CLASSIFIER.md, docs/RASFF_INGESTION.md, docs/LOKSABHA_SAMPLING.md, docs/BACKTEST_SAMPLING.md, docs/PAPER_SCOPING.md.